

“Niñas Atómicas” (Atomic Girls) Hands-On Detector Assembly Instructions

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1 Detector Assembly

1.1 Materials

- Gloves
- Aluminium foil
- Plastic scintillator x2
- Teflon

- MicroFC-60035SMT SiPM x2
- Hexagonal casing x2
- Main detector case
- Optical grease
- Female to Female Dupont Wires x6

1.2 Instructions

1. Wearing gloves, wrap each plastic scintillator in aluminium foil making sure to leave exposed the side that will come in contact with the SiPM. Then, wrap the covered sides of the scintillator in Teflon.



Figure 1: Wrapped plastic scintillator. Left scintillator shows the uncovered side.

2. Place a SiPM in each hexagonal casing of the detector. Ensure that the pins fit into the hole of the casing.

Note: When placing the SiPM, be careful not to touch or damage the silicon.

3. Apply a drop of optical grease to the silicon of the SiPM. Then place each scintillator in its respective hexagonal casing. Ensure that the exposed side of the scintillator is in contact with the SiPM.

Note: In case the scintillator must be replaced, ensure to apply another drop of optical grease to the SiPM before installing the new scintillator.

4. Connect 3 female to female dupont wires to the pins of each SiPM. Remember which wire corresponds to which pin (GND, Vin and Signal).
5. Place one array (SiPM + Scintillator + hexagonal casing + Wires) into the bottom of the detector case. The wires must pass through the slit

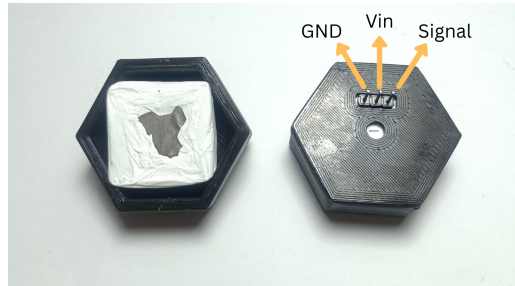


Figure 2: Hexagonal casing with SiPM and plastic scintillator. The uncovered side of the scintillator is in contact with the SiPM.

of the case. Ensure that the covered side of the scintillator is facing upward.

6. Place the other array into the detector case but this time with the covered side of the scintillator facing down. Both scintillators must be in contact via their covered faces in the middle of the detector case. Then place the top cover of the detector.



Figure 3: Assembled detector and its top cover.

2 Electrical Connections

2.1 Materials

- Muon Count PCB
- Arduino UNO
- Flathead screwdriver
- MicroSD Card Module
- MicroSD Card
- Female to Male Dupont Wires x9
- Multimeter

2.2 Instructions

1. Connect the 3 wires of each SiPM to the pins of the Muon Count PCB as shown in the Figure 4. Be careful to connect the *Signal* wire to the pin labeled *S*, the *GND* wire to the pin on the right, and the *Vin* wire to the “+” pin.

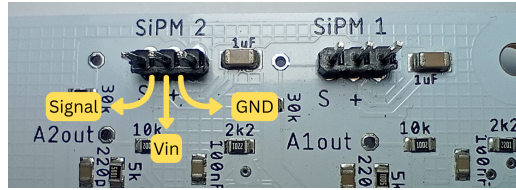


Figure 4: SiPM connection pins on the Muon Count PCB.

2. Connect the Muon Count PCB to the Arduino using 3 female to male dupont wires as shown in Table 1.

PCB	ARDUINO
GND	GND
3.3V	3.3V
TTL out	D3

Table 1: Pins connections between Arduino UNO and Muon Count PCB.

Then, connect the Arduino to the power supply. This can be done by connecting the Arduino UNO to the computer with a USB cable or directly to a power socket with a transformer.

3. After connecting the Arduino UNO to the power supply, it's necessary to calibrate the 3 potentiometers on the Muon Count PCB using a multimeter and a flathead screwdriver.



Figure 5: Arduino UNO connections pins on the Muon Count PCB.

- a) The potentiometers (1) and (2) must be modulated so that the voltage at $TCom1$ and $TCom2$ is 2.18V. Rotate the screw **clockwise** to increase the voltage.
- b) The potentiometer (3) must be modulated so that the voltage at the pin labeled as “29.4V” is 29.4V. Rotate the screw **counter-clockwise** to increase the voltage.

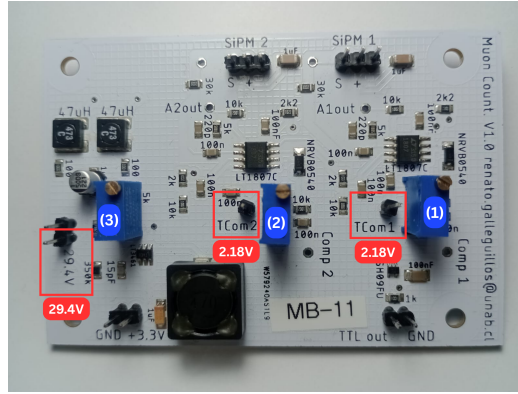


Figure 6: In blue: Muon Count PCB potentiometers. In red: Potentiometer voltage testing pins.

Note: At the moment of measuring the potentiometers voltages, it's important to be careful with the connections of the measuring probes on the multimeter and to take measurements on the correct voltage scale (DC Voltage “V–”). In generics multimeters, the scale for measuring the voltage on $TCom1$ and $TCom2$ should be set to 20V, and to 200V for $TCom3$. See Figure 7 for reference.

4. Finally, the MicroSD Card Module must be connected to the Arduino Uno following the Table 2:

Then, connect the microSD card to the module.

Note: For the correct work of the detector it's necessary to reset the Arduino UNO.



Figure 7: Generic Multimeter. In green: the connections of the measuring probes on the multimeter. In red: The DC voltage scales that should be used.

MicroSD Card Module	Arduino
Vcc	5V
GND	GND
CS	D10
MOSI	D11
MISO	D12
SCK	D13

Table 2: Pins connections between MicroSD Card Module and Arduino Uno.

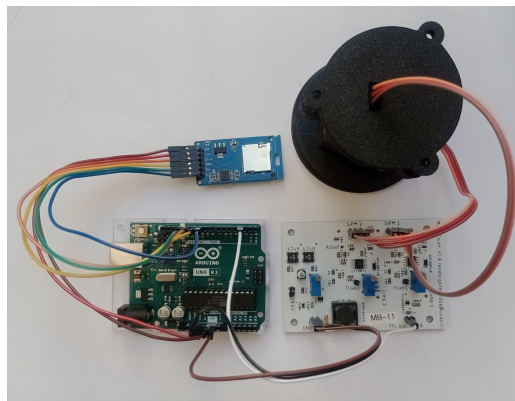


Figure 8: Complete Muon Detector setup.

3 Taking Data

3.1 Monitored Detection

The monitored detection consists in visualizing the muons passing through the detector using the serial monitor of the Arduino IDE software with a computer. If you don't have the software, this can be downloaded for free from this [LINK](#).

For the monitored detection, the Arduino Uno must be connected to the computer with the Arduino IDE software. Then, open the Serial Monitor to visualize the data.

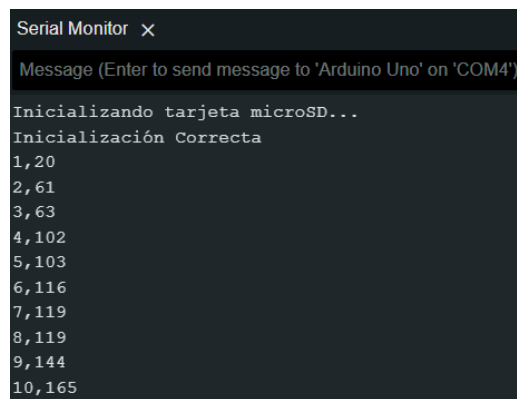


Figure 9: Data on Arduino IDE's serial monitor.

The first number before the comma corresponds to the number of the detected muon, and the next number corresponds to the seconds that have elapsed since the start of the detection.

Note: Before opening the Serial Monitor, make sure to configure the board correctly in the Arduino IDE.

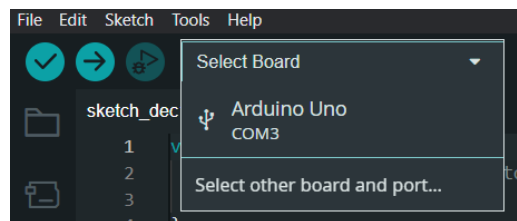


Figure 10: Board selection on Arduino IDE.

At the same time that the Serial Monitor shows the incoming muons, the data is saved to the microSD card in `.txt` format.

3.2 Unmonitored Detection

In the unmonitored detection, the Arduino Uno is not connected to the computer, but is connected to a regular power supply. In this case, real-time data cannot be viewed but it will be saved on the microSD card.

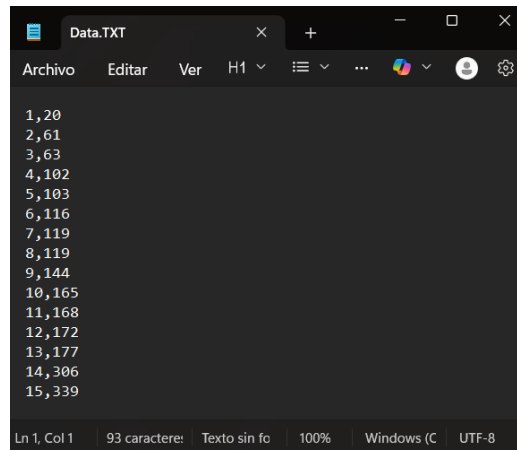


Figure 11: Data on `.txt` file.